

Utility-Enhanced Personalized Privacy Preservation in Hierarchical Federated Learning

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Abstract—Federated learning (FL) is a distributed learning framework that allows clients to jointly train a model by uploading parameter updates rather than sharing local data. FL deployed on a client-edge-cloud hierarchical architecture, named Hierarchical Federated Learning (HFL), can accelerate model training and accommodate more clients with reduced communication cost via edge aggregation. Unfortunately, HFL suffers from privacy risks since the submitted parameters from clients are vulnerable to privacy attacks. To address this issue, we propose a novel Differential Privacy (DP) definition tailored for HFL, i.e., Group Local Differential Privacy (GLDP). We design the Sampling-Randomizing-Shuffling (SRS) mechanism to implement GLDP in HFL, where the sampling process is employed to achieve a stronger level of privacy protection with less noise added. By combining the randomized response and the shuffling mechanism, our proposed SRS mechanism can achieve client-level personalization within ρ_k -GLDP for privacy preservation while balancing model performance and privacy protection in HFL. Privacy analysis and convergence analysis are conducted to provide theoretical performance guarantees. Experimental results based on real-world datasets verify the effectiveness of SRS.

Index Terms—Federated learning, differential privacy, randomized response, shuffling, hierarchical federated learning.

I. INTRODUCTION

FEDERATED Learning (FL), as a distributed machine learning framework, collaboratively trains a joint model by transmitting local updates to a central server instead of sharing their original data, thus preserving data privacy [1]. Initially, FL research primarily revolved around two paradigms: cloud-based FL and edge-based FL, while both of them are impeded by computation cost, communication overhead and network latency. To address these limitations, cloud-edge-end hierarchical federated learning (HFL) [2], [3], [4], [5] was introduced as a great supplement to reduce the network latency and energy consumption in task offloading, which improves the user's service quality [6], [7].

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This hierarchical structure adds complexity to privacy preservation by expanding the attack surface from a central server to multiple edge servers, potentially increasing the risk of privacy leakage. In particular, the model parameters and outputs of the trained model are regarded as the attack surfaces [8], [9], [10], [11]. By analyzing the uploaded parameters of clients, attackers can snoop on their private information [12], [13], [14], [15], [16]. To defend against these attacks, researchers propose a variety of privacy-preserving methods, among which Differential Privacy (DP) is considered as the benchmark for privacy preservation since it provides strong and rigorous privacy guarantees for FL [17], [18], [19]. Some researchers implement DP mechanisms in FL by adding noise with perturbation [20], [21], [22]. However, they usually add excessive noise, resulting in low accuracy of the global model [20], [23]. Additionally, they ignore clients' personalized privacy requirements and set the same privacy budget for all clients [24], [25], which leads to the over-protection of privacy and low model performance. Directly integrating DP into HFL introduces excessive noise at each layer, with the cumulative noise negatively impacting the performance of the global model. HFL-DP proposed in [26] adds noise at both the client and edge server layers, which, however, greatly degrades the model performance; especially, when there are more than 500 clients, the model accuracy drops below 50%. This is because adding too much noise interferes with the model training process. Therefore, it becomes imperative to reach a balance between the model performance and privacy protection.

To fill the gap, we propose a novel DP definition, named *Group Local Differential Privacy (GLDP)*, tailored for HFL. The concept of GLDP stems from the group of clients connected to the same server sharing a localized privacy preference. A straightforward solution to achieve GLDP in HFL would be setting a unified privacy preference for all clients in the same group. However, a low unified privacy setting would fail to meet the stricter requirements of some clients, while a high unified setting would introduce excessive noise, degrading model performance. Additionally, the multi-tiered aggregation structure of HFL further complicates this by increasing the risk of privacy leakage at both the edge and cloud levels. The key challenge, therefore, is to balance strong privacy protection with high model performance by carefully selecting group privacy preferences and implementing customized privacy settings for each client. To overcome this challenge, we design the Sampling-Randomizing-Shuffling (SRS) mechanism, which combines **sampling**, **randomized response**, and **shuffling** mechanisms to achieve a

better balance between privacy and performance in HFL. Our main contributions are summarized as follows:

- We first propose a novel DP definition, named GLDP, which considers varying privacy requirements of clients under different edge servers to achieve privacy personalization and suits the HFL architecture better.
- We design the SRS mechanism to achieve further privacy personalization at client level in HFL. Specifically, the randomized response mechanism satisfies GLDP, achieving a unified privacy preference. The sampling process amplifies the efficacy of privacy preservation to further achieve client-level privacy protection, while the shuffling mechanism complements the sampling process and additionally provides the anonymity of clients.
- The sampling process boasts two facets of effectiveness: first, it utilizes the amplification effect of sampling, adding less noise to achieve the same level of privacy preservation. Second, it prevents the over-protection of some special clients' privacy.
- We theoretically analyze the privacy protection guarantee provided by the SRS mechanism, which uses randomized response to implement GLDP, while the combination of randomized response and sampling ensures client-level privacy protection. Moreover, we theoretically prove that HFL with the SRS mechanism is convergent.
- Experimental results show that our proposed SRS mechanism has high model accuracy while achieving GLDP in HFL, thus balancing the model performance and privacy protection. Our scheme can converge and has better model accuracy than other schemes.

The rest of the paper is organized as follows. The related work about DP for FL and HFL is summarized in Section II. The background knowledge about HFL and DP is presented in Section III. In Section IV, we formulate the problem of privacy protection in HFL, propose a new DP definition for HFL, and then design the SRS mechanism to achieve GLDP. In Section V, we theoretically prove that SRS can satisfy the privacy requirements of all clients. The convergence analysis is provided in Section VI. Experimental results are displayed in Section VII, followed by the conclusion in Section VIII.

II. RELATED WORK

Researchers have proposed various methods to defend against privacy attacks in FL, with DP being widely recognized as the benchmark due to its strong privacy guarantees. McMahan et al. [27] first propose a DP-based FL algorithm to provide formal privacy protection. A parallel work [28] introduces an FL algorithm incorporating DP, which enables privacy-preserving model training with the trade-off between client privacy and model performance. An alternative approach presented in [29] leverages both DP and Secure Multiparty Computation (SMC) to safeguard against differential attacks. Wei et al. [30] develop a DP framework for FL by adding noise to clients' parameters, proving its effectiveness in providing privacy protection with a theoretical convergence bound on the loss function. Girgis et al. [31] introduce an algorithm called CLDP-SGD, which

achieves optimal communication efficiency while maintaining local differential privacy. The application of DP in FL has extended to practical scenarios, such as Internet of Vehicles [32], medical image analysis [33] and health data protection [34].

Liu et al. [2] propose an Hierarchical Federated Learning (HFL) system that integrates clients, edge servers, and cloud infrastructure, leveraging the communication capabilities of edge servers. Yuan et al. [4] design an HFL platform in the Local-Area Network (LAN) to accelerate FL training and save Wide-Area Network (WAN) traffic. However, the HFL architecture still exposes a potential risk of privacy leakage, as evidenced by the inadequate protection of training data privacy against inference attacks [12], [13]. Applying DP in HFL presents additional challenges due to the hierarchical structure that amplifies the impact of noise on the global model's performance. Shi et al. [26] propose an HFL-DP scheme that introduces noise at both the client and edge server layers but suffers from decreased model performance when the number of clients is large. Furthermore, personalized privacy requirements of clients are often overlooked, resulting in excessive privacy preservation that compromises model performance. Balancing privacy protection and model performance is a critical challenge in applying DP to HFL.

To tackle these challenges, we propose Group Local Differential Privacy (GLDP), a novel privacy definition tailored for HFL. Additionally, we introduce the Sampling-Randomizing-Shuffling (SRS) mechanism, which combines sampling, randomized response, and shuffling techniques. This innovative approach aims to strike an optimal balance between privacy and performance in HFL.

III. PRELIMINARIES

A. Hierarchical Federated Learning

Federated Learning (FL) is a collaborative machine learning framework designed to address data privacy concerns [1]. In FL, a set of local clients $\mathcal{N} = \{d_1, \dots, d_i, \dots, d_N\}$ download a global model from a center server, then conduct some epochs of local training using their own data, and upload the model updates to the center server for aggregation.

In particular, we denote ω as the parameter in the learning model. Device $d_i \in \mathcal{N}$ owns its local training dataset $D_i = \{(x_j, y_j)\}_{j=1}^{|D_i|}$, where $|D_i|$ denotes the number of training samples in D_i , x_j denotes the j -th input sample, and y_j denotes the corresponding labeled output of x_j . We use the loss function $f^*(x_j, y_j, \omega)$ (e.g., cross-entropy function) to calculate the prediction error of the model for the j -th sample. The purpose of the learning task is to solve the machine learning parameter ω by minimizing the global loss $F(\omega)$. The cloud server cannot access the local dataset, so it cannot directly calculate the global loss $F(\omega)$. We can average the local loss functions $F_i(\omega)$ in weighted manner to compute $F(\omega)$ as follows:

$$F(\omega) = \frac{1}{|\cup_{i=1}^N D_i|} \sum_{i=1}^N F_i(\omega),$$

where $F_i(\omega) = \frac{1}{|D_i|} \sum_{j \in D_i} f^*(x_j, y_j, \omega)$.

In [2], Liu et al. proposed the concept of Hierarchical Federated Learning (HFL), with a structure of three layers of entities, including the cloud server, a set of edge servers $\mathcal{K} = \{e_1, \dots, e_k, \dots, e_K\}$, and the set of local clients $\mathcal{N}$. The cloud server as the root is connected to multiple edge servers in the middle tier, and each edge server e_k is connected to a set of clients, denoted as $\mathcal{N}_k \subset \mathcal{N}$, organized in a tree structure. Its working process can be described as follows.

The cloud server sends the initial network hyperparameters and random model parameters to edge servers for distributing to clients. For each client $d_i \in \mathcal{N}$, it submits the model parameter ω_i as updates to the connected edge server. Edge server e_k collects the local updates and performs edge aggregation to obtain edge model ω_{e_k} , which will be sent to the cloud server for the global aggregation to get the model parameter ω_c . After aggregation, global model parameters are transmitted down the hierarchy to the clients to start another round of training.

B. Differential Privacy

Differential privacy (DP) is first proposed in [19] and becomes a standard for privacy preservation. It provides a strong privacy guarantee by concealing the privacy disclosure of statistical databases, ensuring that query results remain insensitive to changes in a single record of the dataset.

Definition III.1 (Neighboring Datasets): Two datasets D, D' are defined to be neighboring datasets if $\|D' - D\|_1 \leq 1$, where $\|\cdot\|_1$ is the L_1 norm.

Differential privacy is defined based on the neighboring datasets D and D' , with D' differing from D by only one data record:

Definition III.2 (Differential Privacy): A random algorithm $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$, with domain $\mathcal{D}$ and range $\mathcal{R}$, satisfies ϵ -differential privacy if for any pair of neighboring datasets $D, D' \in \mathcal{D}$, and for all sets $O \subseteq \mathcal{R}$ of possible outputs, we have:

$$\Pr[\mathcal{M}(D) \in O] \leq e^\epsilon \Pr[\mathcal{M}(D') \in O],$$

where ϵ is usually a small positive real number, called privacy budget.

Differential privacy could be achieved by the Laplace mechanism [17] or Gaussian mechanism [19], [35], which adds noise sampled from Laplace distribution or Gaussian distribution. Generally, the smaller the value of ϵ , the higher the level of privacy preservation, and the more noise needs to be added.

IV. PROPOSED MECHANISM

A. Threat Model

In the HFL framework, we assume that clients are honest, while edge servers and the cloud server are all semi-honest [36], which means that they will follow the HFL protocol but try to infer clients' information from the submitted model updates. Model updates may be exposed to privacy risks including inference attack [12], [13], model-inversion attack, and reconstruction attack [14], [16]. The goal of these attacks is to obtain sensitive information about the training data, which can be a violation of the individual privacy for clients.

Moreover, different clients have their own privacy requirements, which can be indicated by different privacy budgets. Such an HFL scenario with various privacy requirements can be formally modeled as follows:

Problem IV.1: In HFL consisting of a set of clients $\mathcal{N}$, a set of edge servers $\mathcal{K}$, and a cloud server, while each client d_i has a privacy requirement to be satisfied, which is specified by the privacy budget ϵ_i , the performance of the global model should also be guaranteed.

B. GLDP

In order to resist a variety of privacy attacks in HFL, we propose a novel privacy definition called Group Local Differential Privacy (GLDP), which sets a unique privacy preference for each edge server to achieve edge-level personalization for privacy protection. The privacy preference of edge server e_k is defined below, which is related to the privacy budgets of all connected clients in $\mathcal{N}_k$.

Definition IV.1 (Privacy Preference): The privacy preference of edge server e_k , denoted by ρ_k , is calculated by $\rho_k = \mathcal{Z}(\dots, \epsilon_i, \dots)$, with $\mathcal{Z}$ being the algorithm transforming the set of privacy budgets $\{\dots, \epsilon_i, \dots\}$ into ρ_k .

Then we introduce the definition of GLDP for each edge server e_k with privacy preference ρ_k .

Definition IV.2 (GLDP): For an edge server e_k and its privacy preference ρ_k , a randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ satisfies ρ_k -GLDP if for every pair of neighboring datasets $D, D' \in \mathcal{D}$ and all sets $O \subseteq \mathcal{R}$ of possible outputs, there exists:

$$\Pr[\mathcal{M}(D) \in O] \leq e^{\rho_k} \Pr[\mathcal{M}(D') \in O].$$

C. SRS Mechanism

To address the challenges of balancing performance and privacy in HFL, we propose the SRS mechanism, which is illustrated in Fig. 1. In this figure, we take client d_i belonging to edge server e_k as an example. The SRS mechanism can be divided into five main stages: preparation, sampling, randomized response, shuffling and aggregation. Sampling amplifies the privacy protection effect, reducing the amount of noise added into local models. Randomized response achieves GLDP, working alongside the sampling process to meet the individual privacy needs of each client. Finally, shuffling further obfuscates the link between individual clients and their updates, providing an additional layer of anonymity. Together, these components enable the SRS mechanism to meet privacy requirements while introducing significantly less noise and thus minimizing the impact on model performance. The specific process of the SRS mechanism is shown in Algorithm 1. In preparation stage, client d_i uploads its privacy budget. Edge server e_k calculates the corresponding privacy preference and transmits it back to the clients. In sampling stage, client d_i samples the dimensions of local updates with a certain sampling rate based on privacy requirements from edge server e_k and the client itself. Then client d_i randomizes the sampled values, ensuring differential

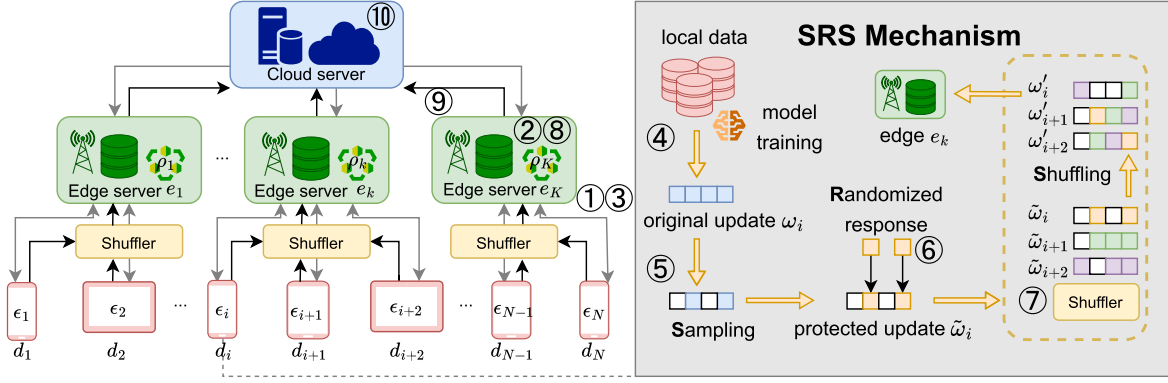


Fig. 1. Illustration of the SRS mechanism deployed in HFL.

privacy of updates.¹ Then it sends the protected update to a shuffler. In shuffling stage, the shuffler shuffles the updates from clients by permutation, then sends them to edge server e_k . In aggregation stage, edge server e_k aggregates updates from shuffler after pre-processing, and submits edge model to the cloud server. The cloud server aggregates edge models and obtains the global model update. The SRS mechanism ensures that all clients meet their individual privacy requirements while introducing significantly less noise and minimizing the impact on model performance. Specifically, for clients with higher privacy requirements, the sampling process amplifies the privacy preservation effect, ensuring their ϵ_i -DP.

1) *Preparation*: ① Local client $d_i \in \mathcal{N}_k$ connects to edge server e_k , with its privacy budget ϵ_i and information entropy E_i . The privacy budget ϵ_i is personalized, reflecting the client's individualized privacy protection requirement. The information entropy E_i can be calculated as $E_i = -\sum_{j=1}^l P_{(i,j)} \log P_{(i,j)}$, where l denotes the number of sample classes and $P_{(i,j)}$ denotes the probability distribution of the j -th ($j \in \{1, 2, \dots, l\}$) class samples on the local dataset of client d_i [41]. We use the information entropy E_i to represent the amount of information. It reflects that the greater the amount of information, the higher level the privacy protection required.

② Edge server e_k calculates its privacy preference $\rho_k = \mathcal{Z}(\dots, \epsilon_i, \dots)$. The algorithm $\mathcal{Z}$ can be modeled as:

$$\rho_k = \mathcal{Z}(\dots, \epsilon_i, \dots) = \frac{n}{\sum_{d_i \in \mathcal{N}_k} (\frac{E_i}{\sum_{d_j \in \mathcal{N}_k} E_j} \times \frac{1}{\epsilon_i})}. \quad (1)$$

In (1), we use the harmonic mean to obtain ρ_k based on the privacy budgets of clients. This is because the harmonic mean is more sensitive to the minimum value. We design this algorithm $\mathcal{Z}$ to describe that the clients with higher privacy requirements, i.e., with smaller privacy budgets, have a greater impact on the calculation of ρ_k . We choose the harmonic mean instead of directly selecting the smallest ϵ_i to obtain a ρ_k that

better fits most clients, thereby reducing the noise that needs to be added and avoiding over-protection for clients with lower privacy requirements.²

However, if all clients conduct ρ_k -GLDP protection scheme, some clients with higher privacy requirements (e.g., client i with its privacy requirement $\epsilon_i < \rho_k$) may not be fully protected, leading to under-protection for them. We address this issue and amplify the privacy preservation effect by employing a sampling step, thus achieving stronger privacy protection.

③ Edge server e_k broadcasts the model and ρ_k to all clients in $\mathcal{N}_k$. ④ Client d_i needs to solve the FL parameter ω_i by iterations. The parameter ω_i characterizes each output value y_j with loss function $f^*(x_j, y_j, \omega_i)$, and the loss function on the dataset of client d_i is $F_i(\omega_i)$. Client d_i performs $I_i(\theta)$ rounds of local iteration to achieve the local accuracy θ , where $I_i(\theta)$ is formulated as $I_i(\theta) = \xi_i \log(\frac{1}{\theta})$, with constant ξ_i depending on the data size and the machine learning task [42]. In practical, at t -th local iteration, client d_i obtains its original local update as:

$$\omega_i^t = \omega_i^{t-1} - \eta \nabla F_i(\omega_i^{t-1}), \quad (2)$$

until $\|\nabla F_i(\omega_i^t)\| \leq \theta \|\nabla F_i(\omega_i^{t-1})\|$, and η is the predefined learning rate. After $I_i(\theta)$ iterations of training, client d_i gets its original update ω_i .

We assume that the original ω_i of client d_i has z dimensions. Namely, $\omega_i = (\omega_{i1}, \dots, \omega_{ij}, \dots, \omega_{iz})$. Client d_i calculates the l_2 norm $c = \|\omega_i\|_2$ and standard basis vector of original update $\mathbf{S}_i = (s_1, \dots, s_j, \dots, s_z) = \frac{1}{c} \cdot \omega_i$.

2) *Sampling*: ⑤ According to the local privacy budget and privacy preference from edge server e_k , client d_i calculates the sampling rate $\beta = \min\{\frac{e^{\epsilon_i}-1}{e^{\rho_k}-1}, 1\}$. Next, client d_i conducts a random sampling with a sampling rate of β to sample each dimension ω_{ij} from the original update ω_i .

3) *Randomized Response*: ⑥ Then, it generates a privacy parameter $f = \frac{e^{\rho_k}+1}{e^{\rho_k}-1}$, depending on the requirement of privacy guarantee. If the dimension of ω_{ij} is sampled, a parameter a would be picked up in range of $(0, \frac{f}{5})$. The value of ω_{ij} would

¹Randomized response, employing randomization to obscure individual answers, is a statistical technique used in survey research to elicit truthful responses on sensitive questions while preserving respondent anonymity [37]. It could also be applied in FL with DP [38], [39], [40].

²Other functions with similar sensitivity to smaller values can also be employed.

Algorithm 1: SRS Mechanism.

Input: Any client d_i : local privacy budget ϵ_i , original model update $\omega_i = (\omega_{i1}, \dots, \omega_{ij}, \dots, \omega_{iz})$, information entropy E_i

Output: Shuffler: perturbed model updates ω'_i

Preparation

Client d_i :

1 sends ϵ_i and E_i to edge server e_k

Edge server e_k :

2 calculates privacy preference ρ_k according to (1)

3 sends ρ_k to each client

Client d_i :

4 $c \leftarrow \|\omega_i\|_2$

5 obtains unit vector $\mathbf{S}_i = (s_1, \dots, s_j, \dots, s_z) \leftarrow \frac{1}{c} \cdot \omega_i$

Sampling

Client d_i :

6 calculates sampling rate $\beta \leftarrow \min \left\{ \frac{e^{\epsilon_i} - 1}{e^{\rho_k} - 1}, 1 \right\}$

7 samples ω_{ij} from ω_i with the probability of β

Randomized Response

8 for $j = 1$ to z do

9 if ω_{ij} is sampled then

10 $f \leftarrow \frac{e^{\rho_k} + 1}{e^{\rho_k} - 1}$

11 randomly picks $a \in (0, \frac{f}{5})$

$$\tilde{\omega}_{ij} \leftarrow \begin{cases} c \cdot s_j + \sqrt{\frac{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}, & \text{w.p. } \frac{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{2f(e^{\rho_k}+1)} \\ c \cdot s_j - \sqrt{\frac{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}, & \text{w.p. } \frac{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{2f(e^{\rho_k}+1)} \end{cases}$$

12 end

13 else

14 $\tilde{\omega}_{ij} \leftarrow 0$

15 end

16 end

17 $\tilde{\omega}_i \leftarrow (\tilde{\omega}_{i1}, \dots, \tilde{\omega}_{ij}, \dots, \tilde{\omega}_{iz})$

18 sends $\tilde{\omega}_i$ to shuffler

Shuffling

Shuffler:

20 collects $\tilde{\omega}_i$ from all clients

21 for $j = 1$ to z do

22 gets perturbed updates

$(\dots, \omega'_{ij}, \dots)^T \leftarrow \pi((\dots, \tilde{\omega}_{ij}, \dots)^T)$

23 end

24 for $d_i \in \mathcal{N}_k$ do

25 $\omega'_i \leftarrow (\omega'_{i1}, \dots, \omega'_{ij}, \dots, \omega'_{iz})$

26 end

27 sends all ω'_i to edge server e_k

be randomized using the following mechanism:

$$\tilde{\omega}_{ij} = \begin{cases} c \cdot s_j + \sqrt{\frac{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}, & \text{w.p. } \frac{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{2f(e^{\rho_k}+1)} \\ c \cdot s_j - \sqrt{\frac{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}, & \text{w.p. } \frac{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{2f(e^{\rho_k}+1)} \end{cases} \quad (3)$$

where c denotes the L_2 norm of original update ω_i , s_j denotes the j -th dimension in standard basis vector $\mathbf{S}_i$, and “w.p.” stands for “with probability”. If the dimension is not sampled, the value of this dimension would be set to 0. The randomized response mechanism achieves ρ_k -GLDP and ensures unbiased results, maintaining the statistical nature of the updates. The design is particularly sensitive to clients with smaller privacy budgets, working in tandem with the sampling process to amplify privacy preservation without adding excessive noise. After sampling and randomizing all dimensions, client d_i sends a new protected $\tilde{\omega}_i$ to the shuffler.

Here, we achieve ρ_k -GLDP by randomized response shown in (3). This randomized response design ensures that the results are unbiased and have controlled variances, with less negative impact on performance. Additionally, this noise is carefully controlled, and the resulting variance is managed to ensure it does not significantly impact the overall convergence of the model. Further discussion in Section VI shows that the model continues to converge toward an optimal solution even with the presence of noise, maintaining both privacy and performance.

Moreover, we utilize the amplification effect of the sampling process on differential privacy. By this method, the sampled information of clients with a smaller privacy budget can meet their privacy requirements ϵ_i . Therefore, we not only provide sufficient protection for the clients with higher privacy requirements in HFL by the sampling process, but also avoid the over-protection for other clients, balancing the model accuracy and personalized privacy protection requirements.

4) *Shuffling*: ⑦ Here we assume that the shuffler is trusted. The inclusion of the shuffler in our mechanism is grounded in common practices in privacy-preserving mechanisms [31], [43], [44]. The shuffler collects all protected updates $\tilde{\omega}_i$ from clients in $\mathcal{N}_k$, thereby further enhancing privacy protection by adding an additional layer of anonymity to clients. Then the shuffler aligns all updates and performs random permutation π on all values of each dimension. Specifically, this permutation is implemented on $(\tilde{\omega}_{1j}, \dots, \tilde{\omega}_{Nj})^T$, and π will be changed every round. Next, the shuffler rebuilds the permuted updates $\omega'_i = (\omega'_{i1}, \dots, \omega'_{ij}, \dots, \omega'_{iz})$, then sends ω'_i to edge server e_k .

Notably, shuffling works in conjunction with sampling to provide comprehensive privacy protection. Without shuffling, the edge server may use statistical analysis to detect the specific sampling rate, thereby compromising clients' privacy. By integrating sampling, randomized response, and shuffling, the SRS mechanism simultaneously ensures anonymity for all clients, achieves the ρ_k -GLDP for the edge server, and meets the higher privacy requirements of some clients.

5) *Aggregation*: ⑧ Edge server e_k receives all rebuilding model parameters ω'_i from the shuffler. For each dimension j of ω'_i , edge server e_k first counts the number of non-zero value t_j . Different from the direct average, the edge server calculates the aggregated update value on dimension j as:

$$\omega_{ekj} = \frac{\sum_{d_i \in \mathcal{N}_k} \omega'_{ij}}{t_j}, \quad (4)$$

Algorithm 2: Edge Aggregation.

Input: Shuffler: all permuted updates $\{\omega'_i\}$
Output: Edge server e_k : aggregated edge model ω_{e_k}

Edge server e_k :

- 1 receives all ω'_i from shuffler
- 2 **for** $j = 1$ to z **do**
- 3 $t_j \leftarrow 0$
- 4 **for** $d_i \in \mathcal{N}_k$ **do**
- 5 **if** $\omega'_{ij} \neq 0$ **then**
- 6 $t_j \leftarrow t_j + 1$
- 7 **end**
- 8 **end**
- 9 $\omega_{e_k j} \leftarrow \frac{\sum_{d_i \in \mathcal{N}_k} \omega'_{ij}}{t_j}$
- 10 **end**
- 11 derives the edge model $\omega_{e_k} \leftarrow (\omega_{e_k 1}, \omega_{e_k 1}, \dots, \omega_{e_k z})$

which means dividing the sum of all updates on this dimension by the number of non-zero value. And the edge aggregated update is $\omega_{e_k} = (\omega_{e_k 1}, \dots, \omega_{e_k z})$. The detail of edge aggregation is shown in Algorithm 2.

After that, edge server e_k broadcasts ω_{e_k} to clients in $\mathcal{N}_k$ for the next round of local model computation. The steps of local model computation, sampling, randomized response, shuffling, and edge model aggregation will be repeated until edge server e_k achieves a certain accuracy γ , which is usually the same for all edge servers. Generally, for a convex machine learning task, the required number of edge iterations can be calculated as $J(\gamma) = \frac{\phi(\log \frac{1}{\gamma})}{1-\theta}$ [3], where ϕ is a constant related to the learning task. Note that this framework also works for non-convex machine learning tasks if we figure out the relation between the model accuracy and the convergence iterations. Since the local updates are protected by the SRS mechanism, the edge model ω_{e_k} satisfies ρ_k -GLDP as well.

⑨ After $J(\gamma)$ times of edge aggregation, each edge server e_k uploads ω_{e_k} to the cloud server for the global aggregation. ⑩ At this step, the cloud server receives the edge models from all the edge servers and aggregates them to derive the global model:

$$\omega_c = \frac{\sum_{k=1}^K |D_{\mathcal{N}_k}| \omega_{e_k}}{|\cup_{k=1}^K D_{\mathcal{N}_k}|}.$$

D. Computational Complexity and Scalability

In this subsection, we discuss the computational complexity of the SRS mechanism and its practical scalability in deployment.

In terms of computational complexity, the SRS mechanism consists of three main components: sampling, randomized response, and shuffling. The sampling process has the complexity $\mathcal{O}(n)$ for n clients, as the selection process can be performed efficiently using probabilistic methods. The randomized response mechanism applies noise to each sampled dimension, which results in a time complexity of $\mathcal{O}(n)$ for n clients. The shuffling step, which anonymizes updates before aggregation, incurs the most significant computational cost, with a complexity of $\mathcal{O}(n \log n)$. This complexity arises because the shuffling

process involves sorting or permuting the updates from all clients to break the link between individual clients and their updates. Efficient sorting algorithms, such as merge sort or quicksort, typically run in $\mathcal{O}(n \log n)$.

Thus, the overall time complexity of our SRS mechanism is $\mathcal{O}(n \log n)$ for n clients, ensuring scalability even in large-scale FL environments. The SRS mechanism demonstrates superior efficiency by introducing less noise into clients' updates. This makes the SRS mechanism well-suited for practical deployment in large-scale HFL systems, balancing strong privacy guarantees with efficient runtime performance.

Deploying the SRS mechanism in large-scale HFL systems could introduce challenges related to scalability, computational resources, and network bandwidth. In real-world deployments, edge servers and clients may have limited resources, requiring careful tuning of the parameter of SRS mechanisms to ensure efficient operations. We further explore the influences of parameters on HFL with the SRS mechanism in Section VII.C.

V. PRIVACY ANALYSIS

In this section, we theoretically demonstrate that SRS can satisfy the privacy requirements of all clients, particularly those with stricter privacy demands than ρ_k -GLDP, as the sampling process can amplify the efficacy of privacy preservation.

Theorem 1: Given a value ω_{ij} with corresponding parameters c, s_j, f , and a . The proposed randomized mechanism $\mathcal{M}$ of (3) provides ρ_k -GLDP.

Proof: According to (3), we have $\tilde{\omega}_{ij} = \mathcal{M}(\omega_{ij})$.

Case 1: If $\tilde{\omega}_{ij} = c \cdot s_j + \sqrt{\frac{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}$, then for any $\omega'_{ij}, \omega''_{ij}$,

$$\begin{aligned} \frac{\Pr[\mathcal{M}(\omega'_{ij}) = \tilde{\omega}_{ij}]}{\Pr[\mathcal{M}(\omega''_{ij}) = \tilde{\omega}_{ij}]} &\leq \frac{\max \Pr[\mathcal{M}(\omega'_{ij}) = \tilde{\omega}_{ij}]}{\min \Pr[\mathcal{M}(\omega''_{ij}) = \tilde{\omega}_{ij}]} \\ &= \frac{\max_a (f-a)(e^{\rho_k}-1) + f(e^{\rho_k}+1)}{\min_a (-f+a)(e^{\rho_k}-1) + f(e^{\rho_k}+1)} \\ &= \frac{f(e^{\rho_k}-1) + f(e^{\rho_k}+1)}{(-f)(e^{\rho_k}-1) + f(e^{\rho_k}+1)} = e^{\rho_k}. \end{aligned}$$

Case 2: If $\tilde{\omega}_{ij} = c \cdot s_j - \sqrt{\frac{(f-a)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}{(a-f)(e^{\rho_k}-1)+f(e^{\rho_k}+1)}}$, then for any $\omega'_{ij}, \omega''_{ij}$,

$$\begin{aligned} \frac{\Pr[\mathcal{M}(\omega'_{ij}) = \tilde{\omega}_{ij}]}{\Pr[\mathcal{M}(\omega''_{ij}) = \tilde{\omega}_{ij}]} &\leq \frac{\max \Pr[\mathcal{M}(\omega'_{ij}) = \tilde{\omega}_{ij}]}{\min \Pr[\mathcal{M}(\omega''_{ij}) = \tilde{\omega}_{ij}]} \\ &= \frac{\max_a (f-a)(e^{\rho_k}-1) + f(e^{\rho_k}+1)}{\min_a (-f+a)(e^{\rho_k}-1) + f(e^{\rho_k}+1)} \\ &= \frac{f(e^{\rho_k}-1) + f(e^{\rho_k}+1)}{(-f)(e^{\rho_k}-1) + f(e^{\rho_k}+1)} = e^{\rho_k}. \end{aligned}$$

So we have $\frac{\Pr[\mathcal{M}(\omega'_{ij}) = \tilde{\omega}_{ij}]}{\Pr[\mathcal{M}(\omega''_{ij}) = \tilde{\omega}_{ij}]} \leq e^{\rho_k}$ holds for any given $\omega'_{ij}, \omega''_{ij}$.

Thus, the proposed mechanism satisfies ρ_k -GLDP. ■

Theorem 2: Given any algorithm $\mathcal{M}$ that satisfies ρ' -GLDP and a sampling process $\mathcal{S}_\beta$ with sampling rate $\beta \in (0, 1)$, algorithm $\mathcal{M}(\mathcal{S}_\beta)$ provides a stronger privacy preservation, ρ'' -GLDP, where $\rho' > \rho''$ and $\rho'' = \ln(1 - \beta + \beta e^{\rho'})$.

Proof: We use $\mathcal{M}(D)$ to denote the algorithm on dataset D . $\mathcal{S}_\beta(D)$ denotes the dataset sampled from D with sampling rate β . Specifically, to clarify the different record in neighboring datasets, here we use D and $D_{-\Upsilon}$ to represent neighboring datasets, where $D_{-\Upsilon}$ denotes the dataset resulted from removing record Υ from D .

For any output O , let $Z = \Pr[\mathcal{M}(\mathcal{S}_\beta(D)) \in O]$ and $X = \Pr[\mathcal{M}(\mathcal{S}_\beta(D_{-\Upsilon})) \in O]$.

We need to show that algorithm $\mathcal{M}(\mathcal{S})$ satisfies ρ'' -GLDP, which is equivalent to show $Z \leq e^{\rho''} X$ and $X \leq e^{\rho''} Z$.

Let T denote a subset of dataset D , we could rewrite Z and X as:

$$\begin{aligned} Z &= \sum_{T \subseteq D} \Pr[\mathcal{S}_\beta(D) = T] \Pr[\mathcal{M}(T) \in O], \\ X &= \sum_{T \subseteq D_{-\Upsilon}} \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T] \Pr[\mathcal{M}(T) \in O]. \end{aligned} \quad (5)$$

Since all the T 's resulted by sampling from D with probability β can be divided into two parts: 1) record Υ is not sampled and 2) record Υ is sampled. According to (5), we have:

$$\begin{aligned} Z &= \sum_{T \subseteq D} \Pr[\mathcal{S}_\beta(D) = T] \Pr[\mathcal{M}(T) \in O] \\ &= \sum_{T \subseteq D_{-\Upsilon}} (1 - \beta) \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T] \Pr[\mathcal{M}(T) \in O] \\ &\quad + \sum_{T_{-\Upsilon} \subseteq D_{-\Upsilon}} \beta \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T_{-\Upsilon}] \Pr[\mathcal{M}(T) \in O]. \end{aligned}$$

We assume $Y = \sum_{T_{-\Upsilon} \subseteq D_{-\Upsilon}} \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T_{-\Upsilon}] \Pr[\mathcal{M}(T) \in O]$, so we get $Z = (1 - \beta)X + \beta Y$.

Let $T' = T_{-\Upsilon}$, we can draw:

$$Y = \sum_{T' \subseteq D_{-\Upsilon}} \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T'] \Pr[\mathcal{M}(T'_{+\Upsilon}) \in O]$$

With an algorithm $\mathcal{M}$ satisfying ρ' -GLDP, we can conclude $\Pr[\mathcal{M}(T'_{+\Upsilon}) \in O] \leq e^{\rho'} \Pr[\mathcal{M}(T') \in O]$ for each T' , O . Hence we have:

$$\begin{aligned} Y &\leq \sum_{T' \subseteq D_{-\Upsilon}} \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T'] e^{\rho'} \Pr[\mathcal{M}(T') \in O] \\ &= e^{\rho'} \sum_{T' \subseteq D_{-\Upsilon}} \Pr[\mathcal{S}_\beta(D_{-\Upsilon}) = T'] \Pr[\mathcal{M}(T') \in O] \\ &= e^{\rho'} X, \\ Z &= (1 - \beta)X + \beta Y \leq (1 - \beta)X + \beta e^{\rho'} X \\ &= (1 - \beta + \beta e^{\rho'}) X = e^{\rho''} X. \end{aligned}$$

To show $X \leq e^{\rho''} Z$, we observe that algorithm $\mathcal{M}$ satisfying ρ' -GLDP implies that $X \leq e^{\rho'} Y$ and hence $Z = (1 - \beta)X + \beta Y \geq (1 - \beta)X + \frac{\beta}{e^{\rho'}} X$. Then we could get:

$$X \leq \frac{1}{1 - \beta + \frac{\beta}{e^{\rho'}}} Z = \frac{e^{\rho'}}{e^{\rho'} - \beta e^{\rho'} + \beta} Z.$$

We now prove that $\frac{e^{\rho'}}{e^{\rho'} - \beta e^{\rho'} + \beta} \leq e^{\rho''} = 1 - \beta + \beta e^{\rho'}$

$$\iff e^{\rho'} \leq (1 - \beta + \beta e^{\rho'})(e^{\rho'} - \beta e^{\rho'} + \beta)$$

$$\iff 0 \leq (\beta - \beta^2)(1 + e^{2\rho'} - 2e^{\rho'}).$$

Since $\beta \in (0, 1)$ and $\rho' > 0$, we have $\beta - \beta^2 > 0$ and $1 + e^{2\rho'} - 2e^{\rho'} > 0$. Hence, $\frac{e^{\rho'}}{e^{\rho'} - \beta e^{\rho'} + \beta} \leq e^{\rho''}$, $X \leq e^{\rho''} Z$. ■

VI. CONVERGENCE ANALYSIS

In this section, we prove the convergence of HFL with the SRS mechanism. Denote that there are B cloud aggregations from the start of training to reaching the global convergence. Between any two consecutive cloud aggregations, we assume that there are m_2 edge aggregations; and there are m_1 local training iterations between every two successive edge aggregations. Denote M as the total number of local training iterations, then we have $M = Bm_1m_2$. We use $[p]$ to represent the edge interval starting from the $(p - 1)m_1$ -th local training iteration to the pm_1 -th local training iteration. We denote $\{q\}$ as the cloud interval from the $(q - 1)m_1m_2$ -th local training iteration to the qm_1m_2 -th local training iteration. Thus, we have $\{q\} = [(q - 1)m_2 + 1] \cup [(q - 1)m_2 + 2] \cup \dots \cup [qm_2]$.

Definition VI.1 (Edge Loss Function): Given a model parameter ω , the edge loss function at edge server e_k is expressed as: $F^k(\omega) = \frac{1}{|D_{N_k}|} \sum_{d_i \in N_k} |D_i| F_i(\omega)$.

Definition VI.2 (Virtually Centralized Gradient Descent Sequence): We use $\omega_c^t, \omega_{e_k}^t$ to represent the global model parameter and edge model parameter at the t -th local training iteration, respectively. We denote $v_{\{q\}}(t)$ as the virtually centralized gradient descent sequence in cloud interval $\{q\}$. The value of $v_{\{q\}}(t)$ synchronizes immediately with ω_c^t after every cloud aggregation as:

$$\begin{aligned} v_{\{q\}}((q - 1)m_1m_2) &= \omega_c^{(q-1)m_1m_2}, \\ v_{\{q\}}(t + 1) &= v_{\{q\}}(t) - \eta \nabla F(v_{\{q\}}(t)). \end{aligned} \quad (6)$$

The non-Independent and Identically Distributed (non-IID) data distribution is a significant attribute of FL. According to [45], the local data distribution can be unbalanced and the model performance can be affected by the non-IID data distribution since the data is generated by each client individually. In this paper, we use a similar measurement method in [46] to measure the two-level non-IID data distribution in a hierarchical system, including the client level and the edge level.

Definition VI.3 (Gradient Divergence): For any parameter ω , the gradient divergence between the local loss function of client d_i , and the edge loss function of edge server e_k is defined as an upper bound of $\|\nabla F_i(\omega) - \nabla F^k(\omega)\|$, denoted as δ_i^k . The gradient divergence between the edge loss function of the k -th edge server and the global loss function is defined as an upperbound of $\|\nabla F^k(\omega) - \nabla F(\omega)\|$, denoted as Δ^k . Specifically,

$$\begin{aligned} \|\nabla F_i(\omega) - \nabla F^k(\omega)\| &\leq \delta_i^k, \\ \|\nabla F^k(\omega) - \nabla F(\omega)\| &\leq \Delta^k. \end{aligned} \quad (7)$$

Furthermore, we define Client-Edge divergence $\delta = \frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \delta_i^k}{|\mathcal{D}|}$, and Edge-Cloud divergence $\Delta = \frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \Delta_i^k}{|\mathcal{D}|}$.

A larger gradient divergence means the dataset distribution is more non-IID. δ reflects the non-IIDness at the client level, while Δ reflects the non-IIDness at the edge level.

Definition VI.4 (Optimal Gap): For an cloud interval $\{q\}$, we define $\chi_{\{q\}}(t) = F(v_{\{q\}}(t)) - F(\omega^*)$, for a fixed $\{q\}$, where t is defined as $t \in [(q-1)\tau, q\tau]$ with $\tau = m_1 m_2$. And ω^* is the optimal model parameter that minimizes $F(\omega)$. According to the convergence lower bound of gradient descent given in Theorem 3.14 of [47], we always have $\chi_{\{q\}}(t) > 0$ for any finite t and $\{q\}$.

Next, we introduce the following three lemmas to provide the basis for subsequent proofs about convergence.

Lemma 1: We assume $F(\cdot)$ is α -continuous (i.e., α -Lipschitz, $\|F_i(\omega) - F_i(\omega')\| \leq \alpha \|\omega - \omega'\|$ for any ω, ω'), β -smooth (i.e., $\|\nabla F_i(\omega) - \nabla F_i(\omega')\| \leq \beta \|\omega - \omega'\|$ for any ω, ω'), and convex. When $\eta \leq \frac{1}{\beta}$, for any cloud interval $\{q\}$, and $t \in [(q-1)\tau, q\tau]$, we have that $\|v_{\{q\}}(t) - \omega^*\|$ does not increase with t , where ω^* is the optimal parameter.

Proof: According to the definition of ω^* , we have

$$\begin{aligned} & \|v_{\{q\}}(t+1) - \omega^*\|^2 \\ &= \|v_{\{q\}}(t) - \eta \nabla F(v_{\{q\}}(t)) - \omega^*\|^2 \\ &= \|v_{\{q\}}(t) - \omega^*\|^2 - 2\eta \nabla F(v_{\{q\}}(t))(v_{\{q\}}(t) - \omega^*) \\ &\quad + \eta^2 \|\nabla F(v_{\{q\}}(t))\|^2. \end{aligned}$$

Because of the β -smoothness of $F(\cdot)$ and Lemma 3.5 from [47], we can have:

$$0 < \chi_{\{q\}}(t) \leq \nabla F(v_{\{q\}}(t))(v_{\{q\}}(t) - \omega^*) - \frac{\|\nabla F(v_{\{q\}}(t))\|^2}{2\beta}.$$

Thus, we can further obtain:

$$-\nabla F(v_{\{q\}}(t))(v_{\{q\}}(t) - \omega^*) < -\frac{\|\nabla F(v_{\{q\}}(t))\|^2}{2\beta}.$$

Therefore,

$$\begin{aligned} & \|v_{\{q\}}(t+1) - \omega^*\|^2 \\ &= \|v_{\{q\}}(t) - \eta \nabla F(v_{\{q\}}(t)) - \omega^*\|^2 \\ &= \|v_{\{q\}}(t) - \omega^*\|^2 - 2\eta \nabla F(v_{\{q\}}(t))(v_{\{q\}}(t) - \omega^*) \\ &\quad + \eta^2 \|\nabla F(v_{\{q\}}(t))\|^2 \\ &< \|v_{\{q\}}(t) - \omega^*\|^2 - \eta \frac{\|\nabla F(v_{\{q\}}(t))\|^2}{\beta} \\ &\quad + \eta^2 \|\nabla F(v_{\{q\}}(t))\|^2 \\ &= \|v_{\{q\}}(t) - \omega^*\|^2 - \eta \left(\frac{1}{\beta} - \eta \right) \|\nabla F(v_{\{q\}}(t))\|^2. \end{aligned}$$

So when $\eta \leq \frac{1}{\beta}$, we have $\|v_{\{q\}}(t+1) - \omega^*\|^2 \leq \|v_{\{q\}}(t) - \omega^*\|^2$. ■

Lemma 2: We assume $F(\cdot)$ is α -continuous, β -smooth, and convex. When $\eta \leq \frac{1}{\beta}$, for cloud interval $\{q\}$, and $t \in [(q-1)\tau, q\tau]$, we have:

1) $\tau, q\tau]$, we have:

$$F(v_{\{q\}}(t+1)) - F(v_{\{q\}}(t)) \leq -\eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla F(v_{\{q\}}(t))\|^2. \quad (8)$$

Proof: Because of the β -smoothness of $F(\cdot)$ and Lemma 3.4 from [47], we have:

$$F(\mathbf{x}) \leq F(\mathbf{y}) + \nabla F(\mathbf{y})^T (\mathbf{x} - \mathbf{y}) + \frac{\beta}{2} \|\mathbf{x} - \mathbf{y}\|^2,$$

for arbitrary $\mathbf{x}$ and $\mathbf{y}$. Thus, we have:

$$\begin{aligned} & F(v_{\{q\}}(t+1)) - F(v_{\{q\}}(t)) \\ &\leq \nabla F(v_{\{q\}}(t))(v_{\{q\}}(t+1) - v_{\{q\}}(t)) \\ &\quad + \frac{\beta}{2} \|v_{\{q\}}(t+1) - v_{\{q\}}(t)\|^2 \\ &\leq -\eta \nabla F(v_{\{q\}}(t))^T \nabla F(v_{\{q\}}(t)) + \frac{\eta^2 \beta}{2} \|\nabla F(v_{\{q\}}(t))\|^2 \\ &= -\eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla F(v_{\{q\}}(t))\|^2. \end{aligned}$$

Lemma 3: For any cloud interval $\{q\}$, when $\eta \leq \frac{1}{\beta}$ and $t \in [(q-1)\tau, q\tau]$, we have:

$$\frac{1}{\chi_{\{q\}}(t+1)} - \frac{1}{\chi_{\{q\}}(t)} \geq \zeta \eta \left(1 - \frac{\eta\beta}{2}\right),$$

where $\zeta = \min_q \frac{1}{\|v_{\{q\}}((q-1)\tau) - \omega^*\|^2}$.

Proof: By definition, we have $\chi_{\{q\}}(t) = F(v_{\{q\}}(t)) - F(\omega^*)$ and $\chi_{\{q\}}(t+1) = F(v_{\{q\}}(t+1)) - F(\omega^*)$. Substituting this into (8), we can obtain:

$$\chi_{\{q\}}(t+1) - \chi_{\{q\}}(t) \leq -\eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla F(v_{\{q\}}(t))\|^2.$$

Equivalently,

$$\chi_{\{q\}}(t+1) \leq \chi_{\{q\}}(t) - \eta \left(1 - \frac{\eta\beta}{2}\right) \|\nabla F(v_{\{q\}}(t))\|^2. \quad (9)$$

According to the convexity condition and the Cauchy-Schwarz inequality, we can draw:

$$\begin{aligned} \chi_{\{q\}}(t) &= F(v_{\{q\}}(t)) - F(\omega^*) \\ &\leq \nabla F(v_{\{q\}}(t))^T (v_{\{q\}}(t) - \omega^*) \\ &\leq \|\nabla F(v_{\{q\}}(t))\| \|v_{\{q\}}(t) - \omega^*\|. \end{aligned}$$

Hence,

$$\frac{\chi_{\{q\}}(t)}{\|v_{\{q\}}(t) - \omega^*\|} \leq \|\nabla F(v_{\{q\}}(t))\|. \quad (10)$$

Substituting (10) into (9), we can get:

$$\chi_{\{q\}}(t+1) \leq \chi_{\{q\}}(t) - \frac{\eta \left(1 - \frac{\eta\beta}{2}\right)}{\|v_{\{q\}}(t) - \omega^*\|} (\chi_{\{q\}}(t))^2.$$

From Lemma 1, we have that for any cloud interval $\{q\}$, and $t \in [(q-1)\tau, q\tau]$, $\|v_{\{q\}}(t) - \omega^*\|$ does not increase with t . So

$\|v_{\{q\}}((q-1)\tau) - \omega^*\| \geq \|v_{\{q\}}(t) - \omega^*\|$. Moreover,

$$-\zeta \geq \frac{-1}{\|v_{\{q\}}((q-1)\tau) - \omega^*\|^2} \geq \frac{-1}{\|v_{\{q\}}(t) - \omega^*\|^2}.$$

Therefore,

$$\begin{aligned} \chi_{\{q\}}(t+1) &\leq \chi_{\{q\}}(t) - \frac{\eta(1 - \frac{\eta\beta}{2})}{\|v_{\{q\}}(t) - \omega^*\|} (\chi_{\{q\}}(t))^2 \\ &\leq \chi_{\{q\}}(t) - \eta\zeta(1 - \frac{\eta\beta}{2}) (\chi_{\{q\}}(t))^2. \end{aligned} \quad (11)$$

Since $\chi_{\{q\}}(t+1)\chi_{\{q\}}(t) > 0$, we are able to divide both sides in (11) by $\chi_{\{q\}}(t+1)\chi_{\{q\}}(t)$:

$$\frac{1}{\chi_{\{q\}}(t)} \leq \frac{1}{\chi_{\{q\}}(t+1)} - \frac{\zeta\eta(1 - \frac{\eta\beta}{2})\chi_{\{q\}}(t)}{\chi_{\{q\}}(t+1)}.$$

From (9) we also know that $\chi_{\{q\}}(t) \geq \chi_{\{q\}}(t+1) > 0$, so $\frac{\chi_{\{q\}}(t)}{\chi_{\{q\}}(t+1)} \geq 1$. Now we can prove that:

$$\frac{1}{\chi_{\{q\}}(t+1)} - \frac{1}{\chi_{\{q\}}(t)} \geq \frac{\zeta\eta(1 - \frac{\eta\beta}{2})\chi_{\{q\}}(t)}{\chi_{\{q\}}(t+1)} \geq \zeta\eta(1 - \frac{\eta\beta}{2}).$$

■

Based on the previous three lemmas, we are now ready to prove that the loss function $F(\cdot)$ is convergent. Our main idea is to show that the parameter ω^t does not deviate much from the virtually centralized sequence $v_{\{q\}}(t)$.

Lemma 4: For any client d_i , assuming $F(\cdot)$ is α -continuous (i.e., α -Lipschitz, $\|F(\omega) - F(\omega')\| \leq \alpha\|\omega - \omega'\|$ for any ω, ω'), β -smooth (i.e., $\|\nabla F(\omega) - \nabla F(\omega')\| \leq \beta\|\omega - \omega'\|$ for any ω, ω'), and convex. We assume the deviation of weights has an upper bound $h(\tau, \delta, \eta)$ and a fixed η . When all the following conditions are satisfied:

- 1) $\eta \leq \frac{1}{\beta}$,
- 2) $\eta\psi - \frac{\alpha h(\tau, \delta, \eta)}{\tau\epsilon} > 0$,
- 3) $F(v_{\{q\}}(q\tau)) - F(\omega^*) \geq \mu$, for $q = 1, \dots, B$,
- 4) $F(\omega_c^B) - F(\omega^*) \geq \mu$, for some $\mu > 0$,

then after $M = B\tau$ local iterations, we have the following convergence upper bound:

$$F(\omega_c^B) - F(\omega^*) \leq \frac{1}{B(\eta\psi - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2})},$$

where $\psi = (1 - \frac{\beta\eta}{2}) \min_q \frac{1}{\|F(v_{\{q\}}((q-1)\tau)) - F(\omega^*)\|}$ and $h(x, \delta, \eta) = \frac{\delta}{\beta}((\eta\beta + 1)^x - 1) - \eta\beta x$.

Proof: Using Lemma 3 and $t \in [(q-1)\tau, q\tau]$, we have:

$$\begin{aligned} &\frac{1}{\chi_{\{q\}}(q\tau)} - \frac{1}{\chi_{\{q\}}((q-1)\tau)} \\ &= \sum_{j=(q-1)\tau}^{q\tau-1} \left(\frac{1}{\chi_{\{q\}}(j+1)} - \frac{1}{\chi_{\{q\}}(j)} \right) \end{aligned}$$

$$\geq \zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right).$$

Summing up the inequality above for all $q = 1, \dots, B$, we obtain:

$$\begin{aligned} \sum_{q=1}^B \left(\frac{1}{\chi_{\{q\}}(q\tau)} - \frac{1}{\chi_{\{q\}}((q-1)\tau)} \right) &\geq \sum_{q=1}^B \zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right) \\ &= B\zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right), \end{aligned}$$

which is equivalent to:

$$\begin{aligned} &\frac{1}{\chi_{\{B\}}(q\tau)} - \frac{1}{\chi_{\{1\}}(0)} - \sum_{q=1}^{B-1} \left(\frac{1}{\chi_{\{q+1\}}(q\tau)} - \frac{1}{\chi_{\{q\}}(q\tau)} \right) \\ &\geq B\zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right). \end{aligned}$$

That is,

$$\begin{aligned} &\frac{1}{\chi_{\{B\}}(q\tau)} - \frac{1}{\chi_{\{1\}}(0)} \\ &\geq B\zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right) + \sum_{q=1}^{B-1} \left(\frac{1}{\chi_{\{q+1\}}(q\tau)} - \frac{1}{\chi_{\{q\}}(q\tau)} \right). \end{aligned} \quad (12)$$

Each term inside the sum symbol in (12) can be expressed as:

$$\begin{aligned} \frac{1}{\chi_{\{q+1\}}(q\tau)} - \frac{1}{\chi_{\{q\}}(q\tau)} &= \frac{\chi_{\{q\}}(q\tau) - \chi_{\{q+1\}}(q\tau)}{\chi_{\{q\}}(q\tau)\chi_{\{q+1\}}(q\tau)} \\ &= \frac{F(v_{\{q\}}(q\tau)) - F(v_{\{q+1\}}(q\tau))}{\chi_{\{q\}}(q\tau)\chi_{\{q+1\}}(q\tau)} \\ &\geq \frac{-\alpha h(\tau, \delta, \eta)}{\chi_{\{q\}}(q\tau)\chi_{\{q+1\}}(q\tau)}. \end{aligned} \quad (13)$$

We assume that $F(v_{\{q\}}(q\tau)) - F(\omega^*) \geq \mu$ for all $\{q\}$. According to Lemma 2, we have $F(v_{\{q\}}(t)) \geq F(v_{\{q\}}(t+1))$, for any $t \in [(q-1)\tau, q\tau]$. So we have $\chi_{\{q\}}(t) = F(v_{\{q\}}(t)) - F(\omega^*) \geq \mu$, for all t and $\{q\}$. Consequently,

$$\frac{-1}{\chi_{\{q\}}(q\tau)\chi_{\{q+1\}}(q\tau)} \geq -\frac{1}{\mu^2}. \quad (14)$$

Combining (13) with (14), the sum in the right-hand side of (12) has a lower bound as:

$$\begin{aligned} \sum_{q=1}^{B-1} \left(\frac{1}{\chi_{\{q+1\}}(q\tau)} - \frac{1}{\chi_{\{q\}}(q\tau)} \right) &\geq -\sum_{q=1}^{B-1} \frac{\alpha h(\tau, \delta, \eta)}{\mu^2} \\ &= (1-B) \frac{\alpha h(\tau, \delta, \eta)}{\mu^2}. \end{aligned} \quad (15)$$

Substituting the above into (12), we get:

$$\frac{1}{\chi_{\{B\}}(q\tau)} - \frac{1}{\chi_{\{1\}}(0)} \geq B\zeta\eta\tau \left(1 - \frac{\eta\beta}{2} \right) + (1-B) \frac{\alpha h(\tau, \delta, \eta)}{\mu^2}. \quad (16)$$

It is also assumed that $F(\omega_c^{B\tau}) - F(\omega^*) \geq \mu$. Similar to (14), we can get:

$$\frac{-1}{(F(\omega_c^{B\tau}) - F(\omega^*))\chi_{\{B\}}(B\tau)} \geq -\frac{1}{\mu^2}.$$

Thus, we have:

$$\begin{aligned} & \frac{1}{F(\omega_c^{B\tau}) - F(\omega^*)} - \frac{1}{\chi_{\{B\}}(B\tau)} \\ &= \frac{\chi_{\{B\}}(B\tau) - (F(\omega_c^{B\tau}) - F(\omega^*))}{(F(\omega_c^{B\tau}) - F(\omega^*))\chi_{\{B\}}(B\tau)} \\ &= \frac{F(v_{\{B\}}(B\tau)) - F(\omega_c^{B\tau})}{(F(\omega_c^{B\tau}) - F(\omega^*))\chi_{\{B\}}(B\tau)} \\ &\geq \frac{-\alpha h(\tau, \delta, \eta)}{(F(\omega_c^{B\tau}) - F(\omega^*))\chi_{\{B\}}(B\tau)} \\ &\geq -\frac{\alpha h(\tau, \delta, \eta)}{\mu^2}. \end{aligned} \quad (17)$$

Summing up (16) and (17), we obtain:

$$\begin{aligned} & \frac{1}{F(\omega_c^{B\tau}) - F(\omega^*)} - \frac{1}{\chi_{\{1\}}(0)} \geq B\tau\zeta\eta \left(1 - \frac{\eta\beta}{2}\right) \\ & - \frac{B\alpha h(\tau, \delta, \eta)}{\mu^2} = B\tau \left(\zeta\eta \left(1 - \frac{\eta\beta}{2}\right) - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2} \right). \end{aligned}$$

Since $\chi_{\{1\}}(0) = F(v_{\{1\}}(0)) - F(\omega^*) > 0$, we note that:

$$\begin{aligned} & \frac{1}{F(\omega_c^{B\tau}) - F(\omega^*)} \geq \frac{1}{F(\omega_c^{B\tau}) - F(\omega^*)} - \frac{1}{\chi_{\{1\}}(0)} \\ & \geq B\tau \left(\zeta\eta \left(1 - \frac{\eta\beta}{2}\right) - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2} \right) > 0. \end{aligned}$$

Considering the assumption $\zeta\eta(1 - \frac{\eta\beta}{2}) - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2} > 0$, we can finally prove that:

$$\begin{aligned} F(\omega_c^{B\tau}) - F(\omega^*) &\leq \frac{1}{B(\zeta\eta(1 - \frac{\eta\beta}{2}) - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2})} \\ &= \frac{1}{B(\eta\psi - \frac{\alpha h(\tau, \delta, \eta)}{\tau\mu^2})}. \end{aligned}$$

Then we prove that HFL with the proposed SRS mechanism converges. The basic idea is to study how the parameter ω^t deviates from the virtually centralized sequence $v_{\{q\}}(t)$ when the parameters vary. The noise introduced by our SRS mechanism is controlled by an increase in δ and Δ in Definition VI.3. This ensures that while the variance of the updates is slightly adjusted, the overall convergence properties remain unaffected. The nature of the noise guarantees that the model continues to converge within acceptable limits, preserving both privacy and performance.

In the following three lemmas, we prove that there is an upper bound for the distributed weights deviation for convex loss functions in HFL with the proposed SRS mechanism.

Lemma 5: If $F_i(\cdot)$ is β -smooth, for any cloud interval $\{q\}$ and $t \in [(q-1)\tau, q\tau]$, we have:

$$\|\tilde{\omega}_{d_i}^t - v_{\{q\}}(t)\| \leq g_{d_i}(t - (q-1)\tau),$$

where we define the function $g_{d_i}(x)$ as:

$$g_{d_i}(x) = \frac{\delta}{\beta}((\eta\beta + 1)^x - 1).$$

Proof: We prove by induction that $\|\omega_{d_i}^t - v_{\{q\}}(t)\| \leq g_{d_i}(t - (q-1)\tau)$ for all $t \in ((q-1)\tau, q\tau]$. When $t = (q-1)\tau$, we know that $\tilde{\omega}_{d_i}^t = v_{\{q\}}(t)$ according to the definition of $v_{\{q\}}(t)$. So we have $\|\omega_{d_i}^t - v_{\{q\}}(t)\| \leq g_{d_i}(0)$. Note that when $t \in [(q-1)\tau, q\tau]$, $\tilde{\omega}_{d_i}^t = \omega_{d_i}^t$ since there is no global aggregation within this interval. Combining with (2), for $t \in ((q-1)\tau, q\tau)$, we have:

$$\tilde{\omega}_{d_i}^t = \tilde{\omega}_{d_i}^{t-1} - \eta \nabla F_i(\tilde{\omega}_{d_i}^{t-1}).$$

For the induction, we assume that $\|\omega_{d_i}^{t-1} - v_{\{q\}}(t-1)\| \leq g_{d_i}(t-1 - (q-1)\tau)$ holds true for some $t \in ((q-1)\tau, q\tau)$. We now show that $\|\omega_{d_i}^t - v_{\{q\}}(t)\| \leq g_{d_i}(t - (q-1)\tau)$ holds for t .

$$\begin{aligned} & \|\omega_{d_i}^t - v_{\{q\}}(t)\| \\ &= \|(\tilde{\omega}_{d_i}^{t-1} - \eta \nabla F_i(\tilde{\omega}_{d_i}^{t-1})) - (v_{\{q\}}(t-1) - \eta \nabla F(v_{\{q\}}(t-1)))\| \\ &= \|(\omega_{d_i}^{t-1} - v_{\{q\}}(t-1)) - \eta[\nabla F_i(\tilde{\omega}_{d_i}^{t-1}) - \nabla F_i(v_{\{q\}}(t-1))] \\ & \quad + \nabla F_i(v_{\{q\}}(t-1)) - \nabla F(v_{\{q\}}(t-1))\| \\ &\leq \|\omega_{d_i}^{t-1} - v_{\{q\}}(t-1)\| \\ & \quad + \eta\|\nabla F_i(\tilde{\omega}_{d_i}^{t-1}) - \nabla F_i(v_{\{q\}}(t-1))\| \\ & \quad + \eta\|\nabla F_i(v_{\{q\}}(t-1)) - \nabla F(v_{\{q\}}(t-1))\| \quad (18) \\ &\leq (\eta\beta + 1)\|\omega_{d_i}^{t-1} - v_{\{q\}}(t-1)\| + \eta\delta \\ &\leq (\eta\beta + 1)g_{d_i}(t-1 - (q-1)\tau) + \eta\delta \\ &= (\eta\beta + 1)\left(\frac{\delta}{\beta}((\eta\beta + 1)^{t-1-(q-1)\tau} - 1)\right) + \eta\delta \\ &= \frac{\delta}{\beta}(\eta\beta + 1)^{t-(q-1)\tau} - \frac{\delta}{\beta}(\eta\beta + 1) + \eta\delta \\ &= \frac{\delta}{\beta}(\eta\beta + 1)^{t-(q-1)\tau} - \frac{\delta}{\beta} \\ &= \frac{\delta}{\beta}((\eta\beta + 1)^{t-(q-1)\tau} - 1) \\ &= g_{d_i}(t - (q-1)\tau). \end{aligned} \quad (19)$$

In particular, (18) applies the triangle inequality. (19) holds because of the β -smoothness of $F_i(\cdot)$ and (7). Using the above induction, we can show that $\|\tilde{\omega}_{d_i}^t - v_{\{q\}}(t)\| \leq g_{d_i}(t - (q-1)\tau)$ for all $t \in [(q-1)\tau, q\tau]$.

In the following lemma, we aim to establish a tighter lower bound for the distance between the global model ω^t and the virtual model $v_{\{q\}}(t)$. By proving that this bound is less than or

equal to $G_c(m_1 m_2, \eta)$, we ensure that the convergence properties of the model are preserved.

Lemma 6: For any client d_i , we assume that $F_i(\cdot)$ is β -smooth and convex. Then for any cloud interval $\{q\}$ with a fixed η and $t \in ((q-1)\tau, q\tau]$, we have:

$$\|\omega^t - v_{\{q\}}(t)\| \leq G_c(m_1 m_2, \eta),$$

where

$$\begin{aligned} G_c(m_1 m_2, \eta) &= h(t - (q-1)m_1 m_2, \Delta, \eta) \\ &\quad + h(t - ((q-1)m_2 + p(t) - 1)m_1, \delta, \eta) \\ &\quad + \frac{m_1}{2}(p^2(t) + p(t) - 2)h(m_1, \delta, \eta), \\ h(x, \delta, \eta) &= \frac{\delta}{\beta}((\eta\beta + 1)^x - 1) - \eta\beta x, \\ p(x) &= \left\lceil \frac{x}{m_1} - (q-1)m_2 \right\rceil. \end{aligned}$$

Proof: From (2) and (4), we have:

$$\omega^t = \omega^{t-1} - \eta \frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \nabla F_i(\tilde{\omega}_{d_i}^{t-1})}{|D_{\mathcal{N}_k}|}. \quad (20)$$

Then for $t \in ((q-1)\tau, q\tau]$, we have:

$$\begin{aligned} \|\omega^t - v_{\{q\}}(t)\| &= \|\omega^{t-1} - \eta \frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \nabla F_i(\tilde{\omega}_{d_i}^{t-1})}{|D_{\mathcal{N}_k}|} - v_{\{q\}}(t-1) \\ &\quad + \eta \nabla F(v_{\{q\}}(t-1))\| \end{aligned} \quad (21)$$

$$\begin{aligned} &= \|\omega^{t-1} - v_{\{q\}}(t-1) \\ &\quad - \eta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \nabla F_i(\tilde{\omega}_{d_i}^{t-1})}{|D_{\mathcal{N}_k}|} - \nabla F(v_{\{q\}}(t-1)) \right)\| \\ &= \|\omega^{t-1} - v_{\{q\}}(t-1) \\ &\quad - \eta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| (\nabla F_i(\tilde{\omega}_{d_i}^{t-1}) - \nabla F_i(v_{\{q\}}(t-1)))}{|D_{\mathcal{N}_k}|} \right)\| \\ &\leq \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| (\|\nabla F_i(\tilde{\omega}_{d_i}^{t-1}) - \nabla F_i(v_{\{q\}}(t-1))\|)}{|D_{\mathcal{N}_k}|} \right) \end{aligned}$$

$$\begin{aligned} &\leq \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta\beta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| (\|\tilde{\omega}_{d_i}^{t-1} - v_{\{q\}}(t-1)\|)}{|D_{\mathcal{N}_k}|} \right) \\ &\leq \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta\beta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| (g_{d_i}(t-1 - (q-1)\tau))}{|D_{\mathcal{N}_k}|} \right) \\ &= \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta\beta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| (\frac{\delta}{\beta}(\eta\beta + 1)^{t-(q-1)\tau} - 1)}{|D_{\mathcal{N}_k}|} \right) \end{aligned} \quad (22)$$

$$\begin{aligned} &= \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta \left(\frac{\sum_{d_i \in \mathcal{N}_k} |D_i| \delta}{|D_{\mathcal{N}_k}|} \right) ((\eta\beta + 1)^{t-(q-1)\tau} - 1) \\ &= \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\quad + \eta\delta((\eta\beta + 1)^{t-(q-1)\tau} - 1). \end{aligned} \quad (23)$$

Among the above inequalities, (21) is drew from (20) and (6). (22) holds true because of the triangle inequality. And from Lemma 5, we can have (23). So we obtain:

$$\begin{aligned} \|\omega^t - v_{\{q\}}(t)\| &\leq \|\omega^{t-1} - v_{\{q\}}(t-1)\| \\ &\leq \eta\delta((\eta\beta + 1)^{t-(q-1)\tau} - 1). \end{aligned} \quad (24)$$

When $t = (q-1)\tau$, we have $\omega^t = v_{\{q\}}(t)$ according to the definition, thus $\|\omega^t - v_{\{q\}}(t)\| = 0$. For $t \in ((q-1)\tau, q\tau]$, by summing up (24) over different values of t , we can get:

$$\begin{aligned} \|\omega^t - v_{\{q\}}(t)\| &= \sum_{i=(q-1)\tau+1}^t \|\omega^i - v_{\{q\}}(i)\| - \|\omega^{i-1} - v_{\{q\}}(i-1)\| \\ &\leq \eta\delta \sum_{i=(q-1)\tau+1}^t ((\eta\beta + 1)^{i-1-(q-1)\tau} - 1) \\ &= \eta\delta \sum_{j=1}^{t-(q-1)\tau} ((\eta\beta + 1)^{j-1} - 1) \\ &= \eta\delta \sum_{j=1}^{t-(q-1)\tau} (\eta\beta + 1)^{j-1} - \eta\delta(t - (q-1)\tau) \\ &= \eta\delta \frac{(1 - (\eta\beta + 1)^{t-(q-1)\tau})}{-\eta\beta} - \eta\delta(t - (q-1)\tau) \\ &= \eta\delta \frac{((\eta\beta + 1)^{t-(q-1)\tau} - 1)}{\eta\beta} - \eta\delta(t - (q-1)\tau) \\ &= \frac{\delta}{\beta} ((\eta\beta + 1)^{t-(q-1)\tau} - 1) - \eta\delta(t - (q-1)\tau) \\ &= h(t - (q-1)\tau). \end{aligned}$$

With the help of the weight deviation upper bound, we are able to prove the convergence of HFL with the SRS mechanism for convex loss functions.

Theorem 3 (Convex): For any client d_i , assuming $F_i(\omega)$ is α -continuous, β -smooth and convex, then after $M = B\tau$ local iterations, we have the following convergence upper bound in HFL:

$$F(\omega_c^B) - F(\omega^*) \leq \frac{1}{B(\eta\psi - \frac{\alpha G_c(m_1 m_2, \eta)}{m_1 m_2 \mu^2})}.$$

Proof: By substituting $h(\tau, \delta, \eta)$ in Lemma 4 with $G_c(m_1 m_2, \eta)$ in Lemma 6, we prove Theorem 3.

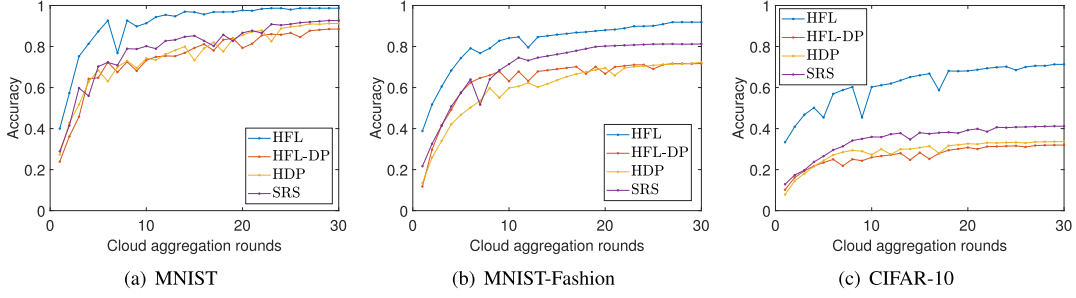


Fig. 2. Accuracy comparison with baseline, HFL-DP, and HDP across three datasets ($\bar{\rho}_q = 0.5$).

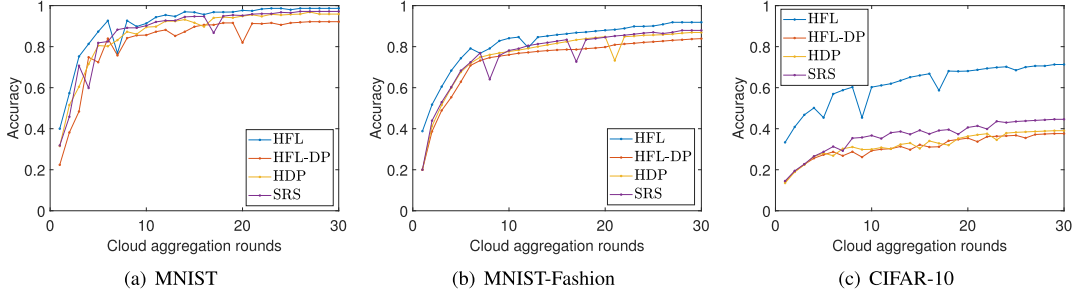


Fig. 3. Accuracy comparison with baseline, HFL-DP, and HDP across three datasets ($\bar{\rho}_q = 1$).

VII. EXPERIMENTAL EVALUATION

A. Experiment Settings

1) *Environment and Model*: We evaluate our proposed SRS mechanism for HFL using Matlab R2021a and Python 3.0 on a GPU-based laptop. We use the Convolutional Neural Network (CNN) model on the MNIST, MNIST-Fashion and CIFAR-10 datasets as the global model in HFL for conducting experiments. The CNNs, consisting of two convolutional layers with ReLU units, are trained with mini-batch Stochastic Gradient Descent (SGD). The cross-entropy loss function is used throughout the training process.

In our experiments, we assume that all clients in the same layer randomly generate their privacy budgets $\epsilon_i \in (0, 10]$, and the communications between different clients are efficient. We set $N = 50$, $K = 10$, $\eta = 0.001$, the average of privacy preference $\bar{\rho}_q = 1$, and the batchsize is 100 except for otherwise specification. Next, we further introduce the datasets and baselines in details.

2) Datasets:

- **MNIST** has 60,000 data samples for training and 10,000 data samples for testing. The samples are 28×28 black and white images of handwritten digits from 0 to 9.
- **MNIST-Fashion** is a dataset of images of clothing and accessories, such as shirts, pants, and shoes. It includes 60,000 training images and 10,000 test images, with each image labeled with one of 10 different clothing categories. The samples are 28×28 black and white images as well.
- **CIFAR-10** consists of 60,000 32×32 color images in 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck), with 6000 images per class. There are 50,000 training images and 10,000 test images.

3) Baselines:

- **HFL** [2] employs a hierarchical structure where clients' updates are first aggregated at edge servers and then forwarded to the central server. We conduct the HFL without any form of DP as an vanilla baseline.
- **HFL-DP** [26] applies Local Differential Privacy (LDP) at both the client and edge server levels in HFL. Noise is added to the client updates before sending them to the edge server and again before sending from the edge to the central server.
- **HDP** [48] introduces a hierarchical level in FL where noise is added at an intermediary level between the clients and the central server.

As for privacy budget of DP, it should be noted that we do not directly generate privacy preferences in a random manner since they are calculated by clients' privacy budgets. Instead, we make the average privacy preference close to a given value (e.g., $\bar{\rho}_q = 1$) by fine-tuning the generated privacy budgets from clients.

B. Accuracy Comparison

We demonstrate the effectiveness of the SRS by comparing the global model accuracy under the same privacy protection level with existing privacy applications on HFL: HFL-DP [26] and HDP [48]. However, unlike our consideration of clients' personalization, the HFL-DP and HDP schemes set the same privacy budget for all clients. To make the comparative experiments more convincing, we set the edge servers' privacy preferences in the SRS the same as the edge servers' privacy budgets in other frameworks (through the fine-tuning mentioned before).

Figs. 2, 3, and 4 present the cases when the privacy preference $\bar{\rho}_q$ is 0.5, 1, and 5, respectively. Note that HFL does not have any privacy protection, so the corresponding blue lines keep the

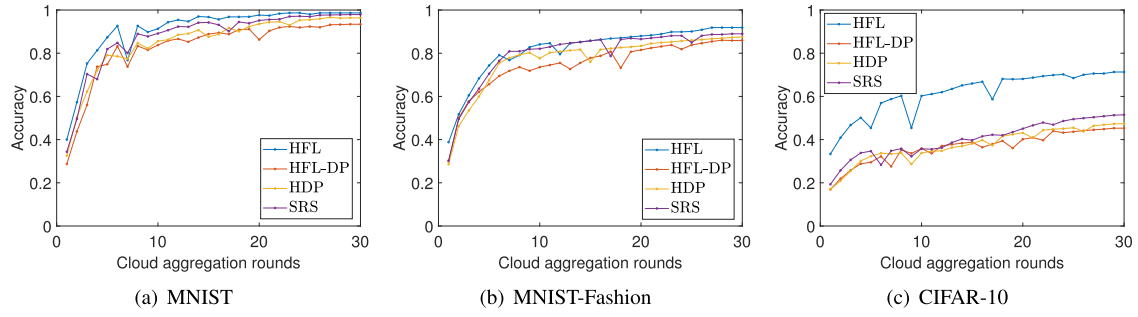


Fig. 4. Accuracy comparison with baseline, HFL-DP, and HDP across three datasets ($\bar{\rho}_q = 5$).

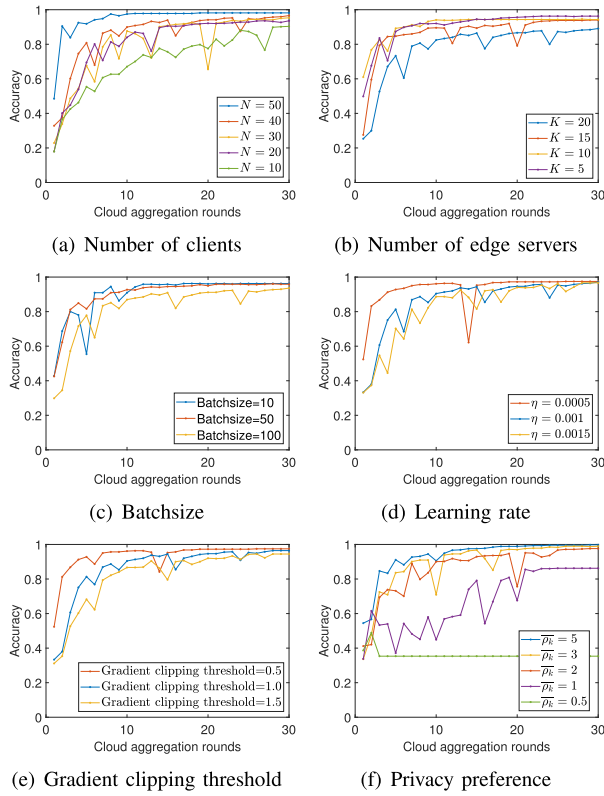


Fig. 5. Evolution of accuracy under different parameter combinations.

same in all figures for a specific dataset. Fig. 2(a) presents the comparison results under MNIST dataset. All lines converge after 30 cloud aggregation rounds. The accuracy of the baseline is higher than all schemes with differential privacy protection. We can also find that the accuracy of the SRS scheme is 0.3% higher than that of the HDP and is 1.5% higher than that of the HFL-DP. The SRS can significantly improve the model accuracy since the excessive noise degrades the accuracies of the HFL-DP and HDP. Fig. 2(b) and (c) show similar results, although the tasks corresponding to these two datasets are more difficult, leading to the lower accuracy.

In Figs. 3 and 4, the changing trend of all lines is similar to that in Fig. 2. When we compare three sets of figures, we conclude that the performance is better when the privacy preference is larger, which is closer to HFL without added privacy

TABLE I
ACCURACY COMPARISON

Dataset	Method	Privacy Preference		
		$\bar{\rho}_q = 0.5$	$\bar{\rho}_q = 1$	$\bar{\rho}_q = 5$
MNIST	HFL	0.987	0.987	0.987
	HFL-DP	0.885	0.923	0.933
	HDP	0.912	0.959	0.964
	SRS	0.926	0.972	0.979
MNIST-F	HFL	0.918	0.918	0.918
	HFL-DP	0.717	0.837	0.859
	HDP	0.723	0.868	0.875
	SRS	0.812	0.879	0.890
CIFAR-10	HFL	0.713	0.713	0.713
	HFL-DP	0.320	0.376	0.453
	HDP	0.336	0.391	0.474
	SRS	0.411	0.446	0.515

protection. In all of the above experiments, the accuracy of our SRS mechanism ends up being higher than the two protection mechanisms HFL-DP and HDP. The improvement of accuracy might appear marginal in Fig. 3(b) and (b). Generally, the SRS mechanism demonstrates a clear advantage over existing schemes in improving model performance.

Additionally, Table I provides a comprehensive comparison of accuracy under various privacy preferences with other methods across the three datasets. This encapsulates the accuracy at convergence as depicted in Figs. 2, 3, and 4. Upon observation, it is evident that our SRS scheme outperforms other existing methods in terms of accuracy.

C. Influences of Parameters on HFL With the SRS

The accuracy of the SRS scheme is affected by several parameters, so we conduct a series of experiments on the MNIST dataset and discussions to investigate how to obtain a better performance.

Fig. 5(a) illustrates how the number of clients affects the global model accuracy. The global accuracy increases as the number of clients increases. This is because the more clients there are, the more updates will be submitted. Although these updates may be affected by noise, overall they have a greater impact on model performance. So when there are more clients, the global model accuracy is higher.

Fig. 5(b) displays the model accuracy with the different number of edge servers and a fixed number of clients $N = 20$. We

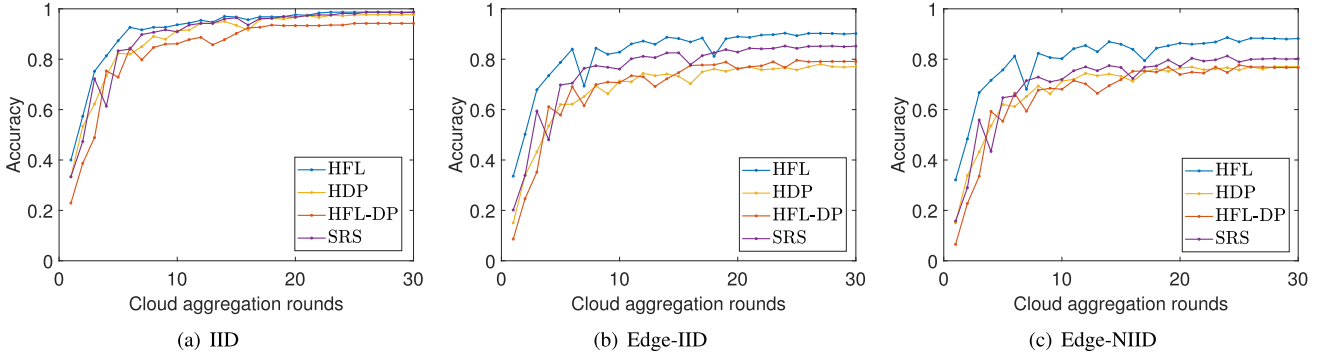


Fig. 6. Accuracy comparison with different non-IID distribution setting on modified MNIST ($\bar{\rho}_q = 1$).

find that the increase in the number of edge servers reduces the global accuracy. The underline reason is that the more edge servers there are, the weaker the role of the edge aggregation. For example, in the extreme case of $K = N = 20$, each edge server is only connected to one client, so the edge aggregation does not work at all. This causes the model to reach convergence late, as the aggregation step that should have occurred at the edge servers is now only implemented at the cloud server. In Fig. 5(b), when the number of edge servers $K = N = 20$, the edge aggregation step is bypassed, leading to a reduction in the total number of local iterations and causing a delay in convergence. For a fair comparison in such cases, we adjust the number of local steps to match the original iteration numbers, ensuring that the total number of local iterations remains constant. Fig. 5(c) illustrates the global accuracy when batchsize is 10, 50, and 100. Specifically, when batchsize is 10, the corresponding blue line is above the other two lines. So we can infer that a small batchsize has a positive effect on achieving high performance.

Fig. 5(d) shows the relationship between the learning rate η and the model accuracy. It can be observed that given other parameters fixed in the MNIST dataset, a lower learning rate makes the model accuracy higher. Fig. 5(e) illustrates the global accuracy when the gradient clipping threshold is 0.5, 1.0, and 1.5. If a gradient value is less than the negative threshold or more than the positive threshold, we set the gradient to the given threshold value (e.g., given gradient clipping threshold is 1.0, if a gradient value is less than -1.0 , it is set to -1.0 and if it is more than 1.0 , then it is set to 1.0). We can find that when the gradient clipping threshold is 0.5, the corresponding red line is above the blue line (gradient clipping threshold is 1.0) and the yellow line (gradient clipping threshold is 1.5). Thus, the smaller gradient clipping threshold is helpful for the higher model accuracy.

Fig. 5(f) presents the effect of the privacy preservation level on the model accuracy. The global model accuracy becomes higher for a bigger $\bar{\rho}_k$. Meanwhile, a greater privacy preference also leads to faster model convergence. In particular, when the privacy preference is very small (e.g., $\bar{\rho}_k = 0.5$), the model will not converge since excessive noise interferes with the training of the model. In order to achieve the balance of privacy protection and model accuracy, privacy preference ρ_k should be chosen wisely. Generally, there is no clear consensus on how to decide

privacy budget in practice, depending on the specific learning tasks and scenarios of experiments, which has always been a challenge in DP research.

D. Discussion About Non-IID

Non-IID local data could significantly influence FL. There are actually two levels of non-IID in HFL: *Edge-IID* and *Edge-NIID* [2].

- **Edge-IID** refers to a non-IID data partition among clients under the same edge server while the distribution of data samples among edge servers is IID.
- **Edge-NIID** witnesses non-IID distribution of clients connected to the same edge server and the data samples among edge servers are also non-IID.

Specifically, in our experiments, we assign each client a single class of training data to achieve a non-IID distribution. For example, with the MNIST dataset, each client is allocated data from one specific digit class.

To discuss whether the above two levels of non-IID impact our proposed SRS scheme, we modify the MNIST dataset to simulate both the Edge-IID and Edge-NIID cases. A series of experiments are conducted on the modified MNIST dataset when the privacy preferences $\bar{\rho}_q = 1$, with the results reported in Fig. 6. Fig. 6(a) shows the results of the control group with IID distribution, which is similar to Fig. 3(a). For the Edge-IID case as shown in Fig. 6(b), we can observe an overall decrease in accuracy for HFL without privacy protection, eventually converging at 90% of accuracy, while the accuracy of HFL-DP and SRS decrease more significantly, but the overall accuracy of our SRS scheme is higher. The accuracy of HFL, HFL-DP, and SRS is further degraded by the Edge-NIID distribution as can be seen in Fig. 6(c). The global model also converges slower, and HFL-DP does not converge after 30 cloud aggregation rounds.

It is worth noting that non-IID conditions also reflect a certain degree of dynamic change of FL, such as random client participation. Our experimental results on non-IID data provide preliminary evidence that the SRS mechanism can adapt to such dynamic changes. By comparing and analyzing the experimental results under these three distributions, we could see that our SRS mechanism works under both two kinds of non-IID distribution

and has a better performance than the existing privacy protection schemes.

VIII. CONCLUSION

In this paper, we introduce GLDP, a novel privacy concept designed for HFL, addressing the diverse privacy requirements of clients. To tackle the challenges in implementing GLDP, we propose the SRS mechanism, which enables personalized privacy at client level. Our SRS mechanism enhances privacy preservation, ensures anonymity, and maintains model accuracy, effectively addressing the challenge of balancing strong privacy guarantees with high model performance in HFL. We also display privacy analysis and convergence analysis to provide theoretical performance guarantees. Experimental results prove that our proposed SRS mechanism achieves better balance between privacy protection and model accuracy than other DP mechanisms in HFL under the same conditions, even in non-IID scenarios.

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